BERNSTEIN
SOCIETE GENERALE GROUP

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Asia Semiconductors & Global Memory

Asia Semis & Global Memory: 3Q26 Earnings Preview - Is AI Still a Long?



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Yes. And we will walk into this earnings season constructively. We find AI is approaching macro "boundary conditions" & are mindful that higher borrowing cost, safety & environmental concerns, etc. may decelerate the growth of semiconductors. Nonetheless, the correction in July should have largely priced these risks in, and the strong fundamental signals & continued shortage should prompt companies to update investors positively in this earnings season.

We see upside risks to our TSMC projection. 3Q26 revenue beat consensus and positive early iPhone sales is supportive for 4Q26 too. Into 1Q27 the benefit of price hike (HSD to low-teens% for adv nodes & MSD% for mature ones) will begin. And with CPUs being the most recent demand upside surprise, we model accelerators (XPU) to contribute advanced packaging revenue additionally but CPUs will contribute as much wafer revenue as XPUs (both mid-teens% of TSMC's total revenue) in 2027. Capacity allocation-wise, we expect incrementally more toward to CPU (vs. XPU) and to ASIC (vs merchant chips). Strong demand justifies high capex, but we don't expect TSMC to update its capex guidance until early 2027.

Capital returns & LTAs should drive memory stocks up. With price increase continuing, but at a moderating pace, offset by negative surprise like employee bonus and slower HBM4 progress, meaningful earnings revisions are becoming increasingly difficult, but we expect capital return & earnings sustainability through LTAs to expand valuation gradually. We prefer Samsung on its HBM4 progress & lower valuation and also CXMT on localization demand mitigating downturn risks & attractive valuation among Chinese peers.

MediaTek is our Top Pick and the recent pullback provides a good setup. For TPU v8 we believe volume shipment commenced a few weeks ago & for v9 the yield progress of substrate suppliers is encouraging. MediaTek may face more competition at v10 but we assume a lower multiple and still see upsides to the stock. Memory weighs on smartphone but share gain should mitigate that. We expect the updates from the earnings call as a positive catalyst, esp. given the recent pullback.

For tier-2 foundries, we turn less negative on UMC, but the expectation is too high & we reiterate Underperform. We're positive on Vanguard but maintain Market-Perform on full valuation. On Novatek, it benefits from Edge AI, but the margin benefit from low-end memory inventory is transitory. Maintain Market-Perform.

BERNSTEIN TICKER TABLE

Ticker	Rating	Cur	8 Oct 2026 Closing Price	Price Target	TTM Rel. Perf.	Reported EPS Cur	2025A	2026E	2027E	Reported P/E (x) 2025A	2026E	2027E
2303.TT (UMC)	U	TWD	147.50	**103.00**	206.6%	TWD	3.34	**7.50**	**6.45**	4.9	3.8	3.6
OLD				88.00				4.58	5.56			
UMC (UMC)	U	USD	23.31	**16.10**	192.6%	USD	0.82	**1.19**	**0.79**	5.7	5.3	5.0
OLD				13.60				0.74	0.89			
5347.TT (Vanguard)	M	TWD	186.50	**194.00**	59.5%	TWD	4.30	**5.93**	**7.76**	0.0	0.0	(0.0)
OLD				146.00				5.70	6.59			
3034.TT (Novatek)	M	TWD	531.00	**490.00**	3.4%	TWD	26.87	**34.32**	**34.03**	19.8	15.5	15.6
OLD				480.00				30.43	32.54			
2330.TT (TSMC)	O	TWD	2,550.00	3,300.00	58.6%	TWD	66.25	110.70	144.85	12.2	8.7	6.5
TSM (TSMC)	O	USD	472.20	554.00	38.9%	USD	10.28	17.18	22.48	14.6	10.4	7.8
2454.TT (MediaTek)	O	TWD	4,690.00	5,500.00	229.7%	TWD	66.17	68.52	169.35	70.9	68.4	27.7
005930.KS (SEC- Samsung)	O	KRW	262,000	440,000	172.8%	KRW	6,611.53	45,072	74,602	39.6	5.8	3.5
005935.KS (SEC-Pref - Samsung)	O	KRW	194,000	374,000	155.6%	KRW	6,611.53	45,072	74,602	29.3	4.3	2.6
SMSN.LI (Samsung)	O	USD	5,020.00	7,857.00	170.2%	USD	116.15	781.29	1,332.17	43.2	6.4	3.8
000660.KS (SK hynix)	O	KRW	1,681,000	2,700,000	303.5%	KRW	60,341	353,627	554,848	27.9	4.8	3.0
MU (Micron)	O	USD	1,088.00	1,300.00	437.4%	USD	8.29	73.78	173.33	131.3	14.7	6.3
285A.JP (KIOXIA)	U	JPY	17,900	49,000	791.7%	JPY	1,014.00	10,485	14,140	17.7	1.7	1.3
688825.CH (CXMT)	O	CNY	50.52	70.00	NA	CNY	0.03	4.04	6.82	N/M	12.5	7.4
ASIAX			1,955.93									
SPX			7,801.77									
EM			1,820.30									

PRICE TARGET CHANGE / ESTIMATE CHANGE IN BOLD

O - Outperform, M - Market-Perform, U - Underperform, NR - Not Rated, CS - Coverage Suspended

MU estimate is Adjusted EPS; 2303.TT, UMC, 5347.TT, 2330.TT, TSM valuation is P/B (x); MU valuation is Adjusted P/E (x);

Source: Bloomberg, Bernstein estimates and analysis.

INVESTMENT IMPLICATIONS

TSMC: We rate TSMC **Outperform** with TP=NT$3,300.00.

UMC: We rate UMC **Underperform** with TP raised to NT$103.00 from NT$88.00 based on the same 2.5x target forward P/B and revised up Q5-Q8 BVPS on higher utilization rate and ASP assumptions

Vanguard: We rate Vanguard **Market-Perform** with TP raised to NT$194.00 from NT$146.00 based on revised up 5x target forward P/B (previously 4x) and revised up Q5-Q8 BVPS mainly on higher visibility for VSMC.

MediaTek: We rate MediaTek **Outperform** with TP=NT$5,500.00.

Novatek: We rate Novatek **Market-Perform** with TP raised to NT$490.00 from NT$480 based on the same 14x target P/E and revised up Q5-Q8 EPS mainly on better growth in SoC segment.

Samsung Electronics: We rate Samsung Electronics **Outperform** with price target of KRW440,000.

SK hynix: We rate SK hynix **Outperform** with target price to KRW 2,700,000.

Micron: We rate Micron **Outperform** with price target of US$1,300.00.

CXMT: We rate CXMT **Outperform** with price target of CNY70.00.

KIOXIA: We rate KIOXIA **Underperform** with price target of JPY49,000.

DETAILS

We expect constructive updates from companies in the coming earnings season.

- More & more signals reveal the AI investments are now so massive & are approaching macro "boundary conditions" that can't be explained simply by the supply vs. demand of semiconductor & other components. Rising borrowing cost is a notable example. Safety & environmental concerns and hence possible regulatory interventions is another one. Any of these, when hitting the boundary, can possibly trigger a deceleration in semiconductor demand, so that other parts of AI investments, and the society overall, have time to catch up.

- On the other hand, all the signals from the upstream semiconductor supply chain still show unabated AI demand and continued shortage. We agree some cost inflation is still in the process of propagating downward & may therefore destroy more demand later. Non-AI demand is more vulnerable, but further damage should relatively small as downstream customers should have made necessary preparations to cope with the cost inflation. More importantly AI demand remains strong, and even when these customers by now are aware of the cost inflation, & particularly the memory price surge.

- Considering both positive & negative signals above, on balance we still decide to position positively into this earnings season, as some of the concerns above should have been priced in in the recent correction. We expect upstream companies to provide some comfort to investors through demand sustainability even after cost inflation, & downstream companies too through broadening AI adoption & rapid usage increase.

With strong demand & disciplined capacity expansion & price increases, TSMC will remain the most trustworthy compounder. We see some upside risks to our projection.

- Demand still broadly surpasses supply at TSMC. Mature-node demand improves but the shortage is particularly acute at leading nodes & advanced packaging. Near term, 3Q26 revenue beat (***link***) & positive initial sales data of iPhone 18 indicated high-end smartphone demand remains resilient despite high memory price. Considering these & capacity constraint, we model TSMC's 2026 full-year revenue to rise by 41% but see a small upside risk to it now. This implies 4Q26 revenue should grow QoQ by c. 10% or more.

- For 2027 customers have accepted the price hike, as the size of the hike, HSD (High Single-Digit) to low-teens% for leading nodes & MSD (Mid-Single Digit) % for mature nodes, is widely considered modest. The effect of the hike will begin to show up in TSMC's financial performance in 1H27 and stay there throughout 2027.

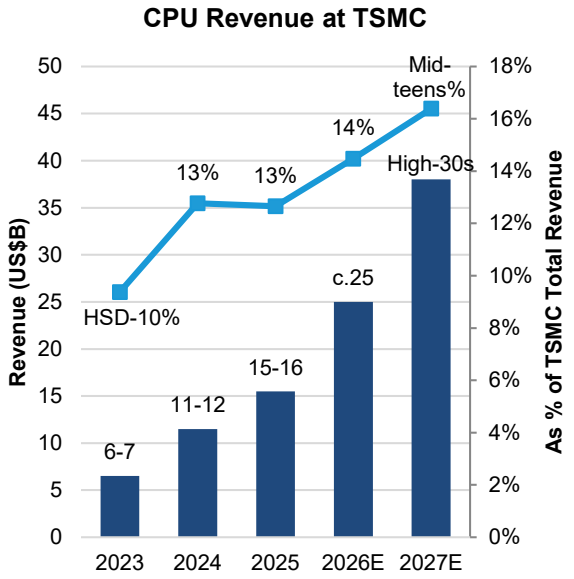
- Agentic workloads make CPUs the most recent demand upside surprise to TSMC (***link***). AI accelerators (XPUs) contribute packaging revenue at TSMC too, but CPUs are getting more incremental wafer capacity. Hence, in terms of wafer revenue we expect CPUs & XPUs will be on par next year, and both will contribute mid-teens% of TSMC's total revenue (Exhibit 1 - Exhibit 2).

- Our recent trip to SEMICON Taiwan also made us conclude that between merchant chips (i.e. ASSP) & ASICs, ASICs are getting more capacity allocation. This is true for XPUs, and also CPUs, with the custom-made CPUs for Google & Amazon (GOOGL.US and AMZN.US, both covered by Bernstein US Internet team) getting notable allocations for example.

- And as most CPUs are packaged with conventional packaging, TSMC's recent capex increase is thus leaning more toward front-end instead of CoWoS capacity to begin with. Then as capacity expansion far exceeds the budget planned in the pipeline, clean rooms is now the biggest bottleneck for TSMC. CPUs needing wafer capacity much more than packaging, as well as OSATs being able to offer CoWoS-equivalent capacity, further makes TSMC allocate available clean rooms more on wafer fabs. We hence forecast Intel (INTC.US, covered by Bernstein US Semi team), through its EMIB-T technology, & OSATs will play an increasing role in the incremental global advanced packaging capacity going forward (Exhibit 3).

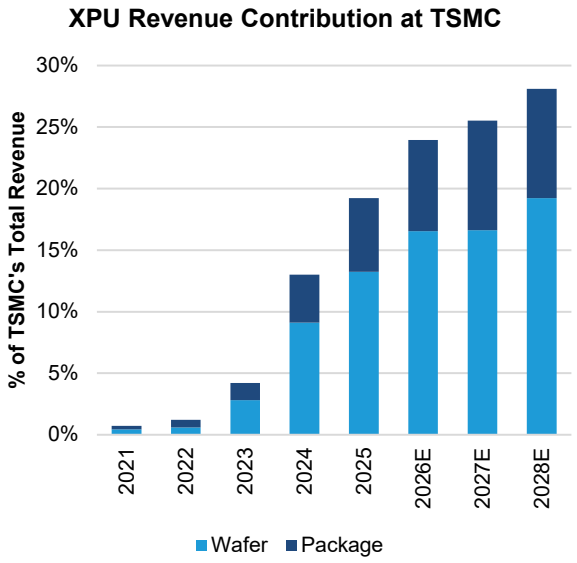
- We expect TSMC's capex to rise higher into 2027 (Exhibit 4), but as we're approaching the year-end we assume TSMC may choose not to modify its 2026 capex guide next week.

EXHIBIT 1: **We estimate CPU-related revenue will reach US$ high-30s billion, accounting for mid-teens% of TSMC's total revenue by 2027.**



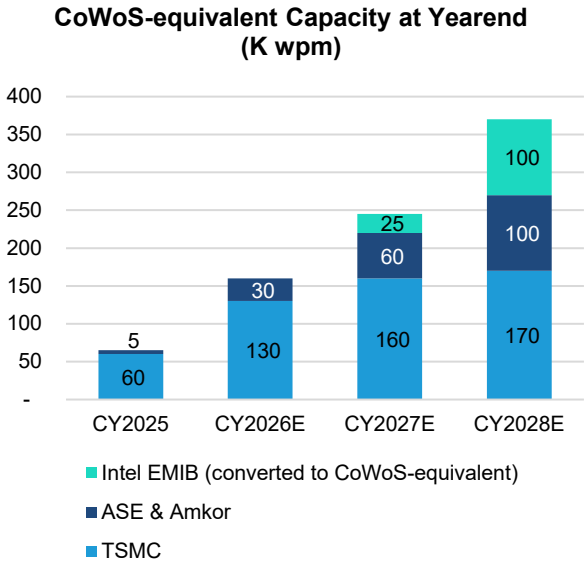
Source: Bernstein estimates and analysis

EXHIBIT 2: **We project CPUs will be about as big as XPUs in terms of wafer revenue contribution in 2027.**



Source: Bernstein estimates and analysis

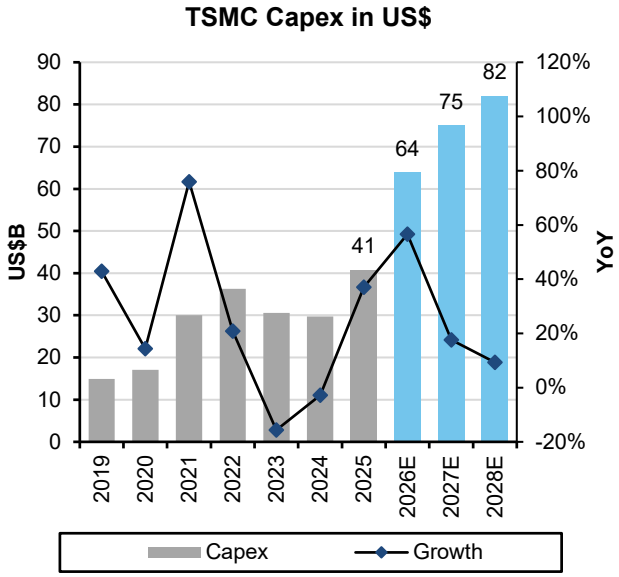
EXHIBIT 3: **We believe TSMC priortizes clean rooms for front-end vs. advanced packaging service impllies others will contribute more incremental advanced packaging capacity expansion.**



Intel's EMIB-T capacity has been converted from panel-based (EMIB-T) to the equivalent CoWoS capacity measured in wafers in this chart.

Both of Amkor (AMKR.US) and ASE (3711.TT) are not covered by Bernstein.
Source: Bernstein estimates and analysis

EXHIBIT 4: **We project TSMC's capex to rise to U$64B this year, and US$75B and US$82B in 2027 & 2028, respectively.**



Source: Company reports, Bernstein estimates and analysis

Given the recent correction but continued price increases & more capital returns, we also remain constructive in memory. We also initiated on CXMT recently & rate it Outperform.

- Recent 3rd-party forecast of 4Q26 DRAM price up 15-20% QoQ, and NAND up 15-20% is slightly ahead of our expectations (***link***). MU's guidance confirmed continued but moderating price increases. While this aligns well with the broad shortage investors are seeing, the surprise came from delays of HBM4, and likely smaller-than-expected HBM price hike for 2027. Employee bonus can also be a negative earnings surprise.

- With both positive & negative surprises, we believe going forward earnings revision is difficult to drive the memory stocks, but capital returns & LTAs will expand multiple and thus drive stock prices up gradually. There are also more & more evidences suggesting Samsung is emerging as a leader, if not already the leader, of HBM4 (***MEMORY TRACKER (Sep): ~20% price hike QoQ,in 3Q, & perhaps teens% in 4Q***, ***Global Memory: Cycle is still elevated, but beware of the long-term threat from China***).

- We initiated with Outperform rating on CXMT on the 21st, Sep (***China Next Winners: CXMT - Redrawing the Memory Map - Initiating with Outperform***). CXMT benefits from China's localization policy, and the company also executes very well to defy the headwind of export restrictions. China's position in memory will affect AI and many other industries. We suggest all tech investors to read our initiation even if they can't buy the stock.

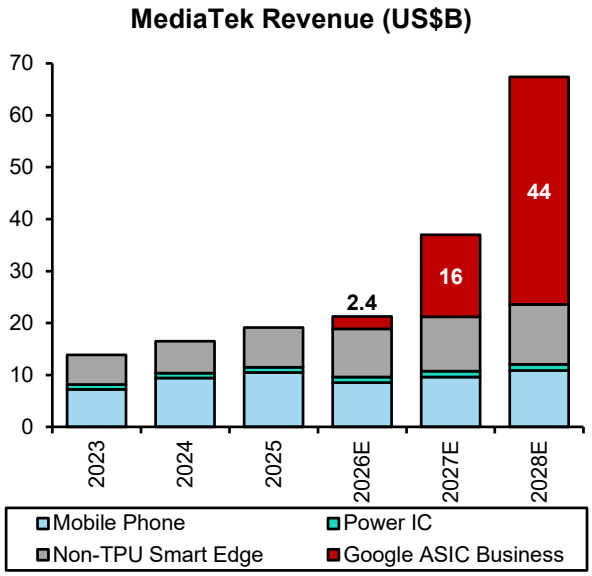
MediaTek is our Top Pick even after we consider possible competition in 2029 & assume lower valuation. The setup into the upcoming earnings release also presents a good opportunity.

- Competition for Google's TPU v10 generation may intensify, and we hence assume a lower multiple and still see significant upside to the stock. This is because we expect upward earnings revisions for the v9 project, as the most critical supply - substrate - is progressing well in production yield. V8 volume shipment, we believe, started not long ago, and we expect smooth execution and sufficient capacity into 2027. Exhibit 5 - Exhibit 7 summarize our forecast & we are notably above consensus. Please find details in our recent reports (***MediaTek: Sizing TAM of ASIC & Upside to MediaTek; raising price target to NT$2,250***, ***MediaTek: ASIC is changing quickly - 4 updates to stay abreast***).

- Additionally, we believe MediaTek is gaining share in smartphone SoC market (Exhibit 8, ***tracker***) & the share gain will mitigate the damage from high memory price.

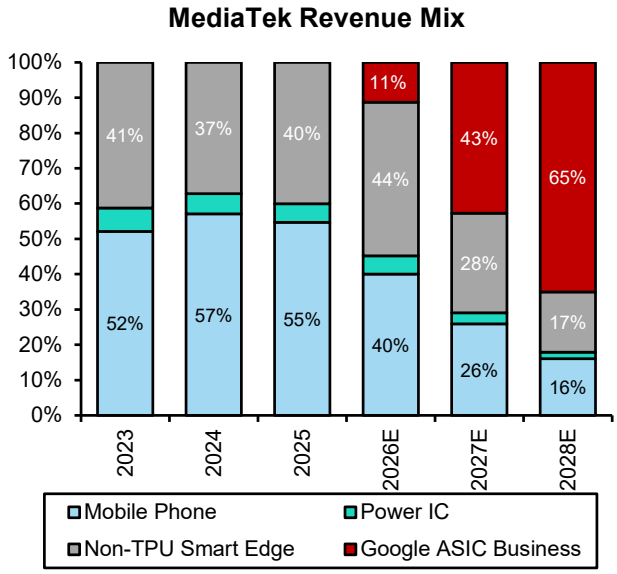
- We hence expect MediaTek to report positively in the upcoming earnings call. The recent pullback provides a good setup for the coming earnings report to surprise investors on the upside. Overall MediaTek is the Top Pick of our coverage.

EXHIBIT 5: **We expect MediaTek's ASIC business to generate US$2B/16B/44B revenue in 2026/27/28, respectively...**



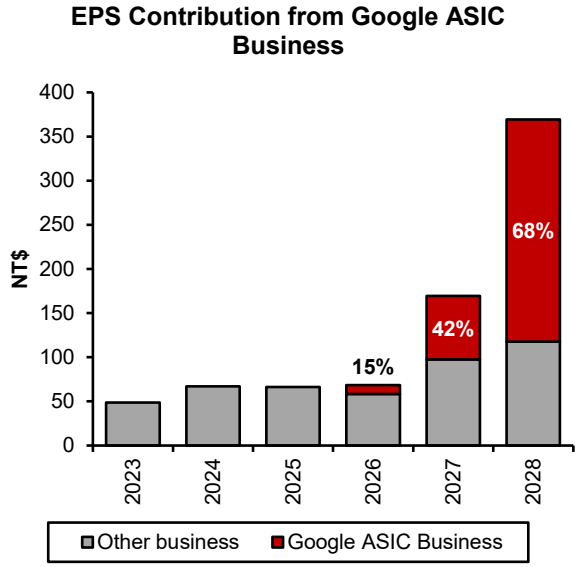
Source: Company reports, Bernstein estimates and analysis

EXHIBIT 6: **...contributing to 11%, 43%, and 65% of its topline...**



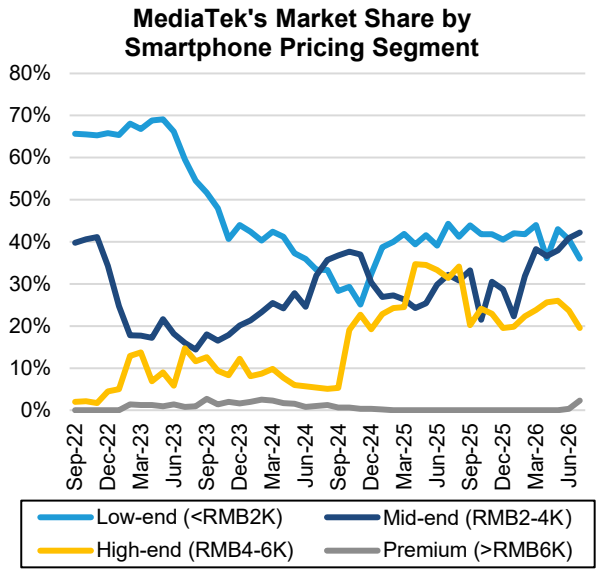
Source: Company reports, Bernstein estimates and analysis

EXHIBIT 7: **... and 15%, 42%, and 68% of its bottom line.**



Source: Company reports, Bernstein estimates and analysis

EXHIBIT 8: **MediaTek has been gaining unit share in mid-end smartphone market YTD.**



Source: CINNO and Bernstein analysis

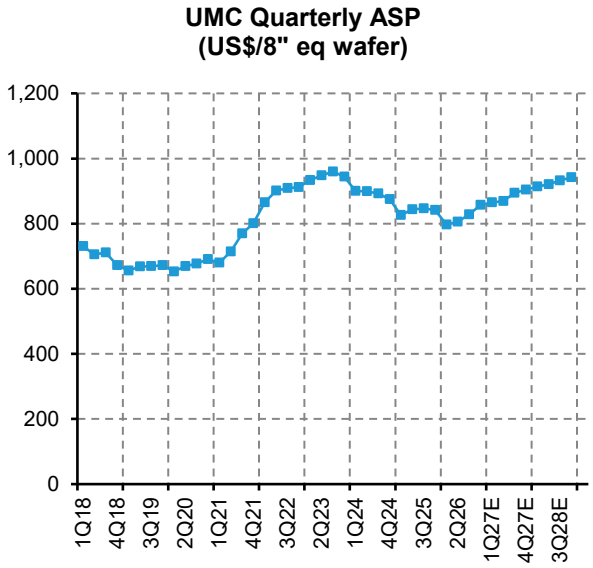
We turn less negative on UMC, but the expectation is too high & we reiterate Underperform.

- We turn less negative on UMC mainly on improving demand for mature nodes amid the AI upcycle. Peripheral chips for connectivity & power management are seeing tight supply, and DDIC orders transferred from Samsung provides an additional windfall. We hence lifted the utilization and pricing assumptions for 2027 & 2028. We now forecast UMC's ASP to increase by 7% in 2027 and 5% in 2028 (Exhibit 9 - Exhibit 10). As a result, we forecast UMC to sustain mid-to-high teens % revenue growth in the medium term (Exhibit 11). Even with heavier depreciation (due to revised up capex plan), margin

expansion should continue (Exhibit 12), supporting ROE improvement through 2026-2028. That said, we expect ROE to remain well below the peak of 28% that took place amid the COVID pandemic (Exhibit 14).

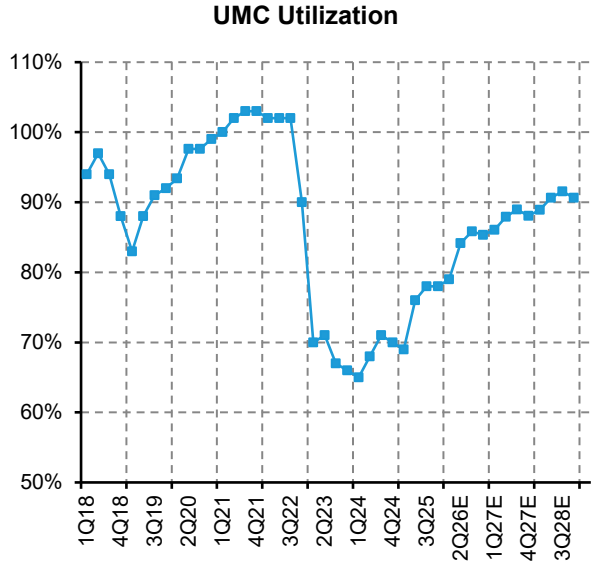
- Our projection is still broadly below consensus for 2027 & 28 likely because we are more conservative on the timing/size of contribution from new business initiatives, such as silicon photonics, DTC (Deep Trench Capacitor), and advanced packaging (Exhibit 13). We set our price target at NT$103.00, up from NT$88.00 based on the same 2.5x target P/B and revised up Q5-Q8 BVPS. This implies 30% potential downside as of October 8 market close. Dividend yield could provide c. 2% return. Maintain Underperform (Exhibit 15).

EXHIBIT 9: **We revised down ASP estimates for 2H26 to reflect more gradual price adjustment, but lifted that for 2027 & 28 as we see price hikes become more promising.**



Source: Company reports, Bernstein estimates and analysis

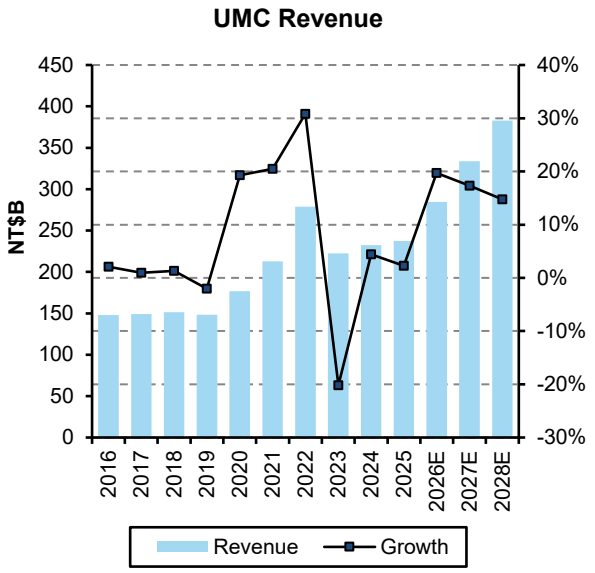
EXHIBIT 10: **We revised up our utilization rate for 2027 & 28 to reflect better- than-expected demand outlook.**



Source: Company reports, Bernstein estimates and analysis

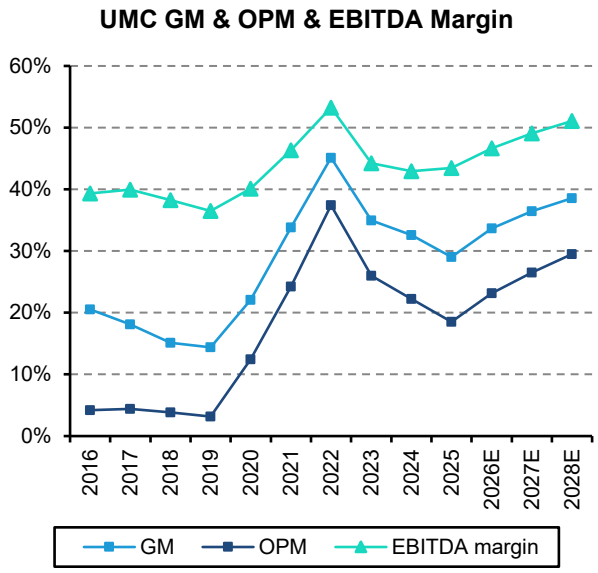
环球研报直通车 一手研报库 订阅报告-微信：waiziyanbao

EXHIBIT 11: **We forecast UMC's revenue to grow by 17% in 2027 and 15% in 2028.**



Source: Company report, Bernstein estimates and analysis

EXHIBIT 12: **We expect UMC's GM & OPM to rise but below the peadk during COVID in 2022.**



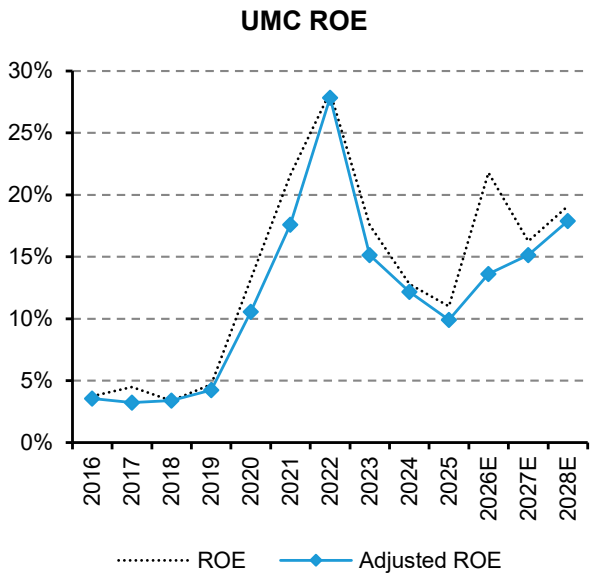
Source: Company report, Bernstein estimates and analysis

EXHIBIT 13: **We're broadly below consensus for 2027 & 2028 revenue and earnings likely because we're more conservative on the timing/size of new business contribution.**

	2026E					2027E					2028E				
UMC	Bernstein New	Old	% Chg	Consensus	% Diff	Bernstein New	Old	% Chg	Consensus	% Diff	Bernstein New	Old	% Chg	Consensus	% Diff
Revenue (NT$ B)	284.4	274.8	3.5%	283.4	0.4%	333.8	315.9	5.7%	350.6	-4.8%	382.9	333.4	14.8%	402.5	-4.9%
YoY	*19.7%*	*15.7%*		*20.3%*		*17.3%*	*14.9%*		*23.7%*		*14.7%*	*5.6%*		*14.8%*	
Gross Margin	33.7%	31.5%	2.2%	33.4%	0.3%	36.5%	33.2%	3.3%	37.7%	-1.3%	38.6%	36.7%	1.8%	39.7%	-1.2%
Operating Margin	23.2%	21.1%	2.1%	23.1%	0.1%	26.5%	23.1%	3.4%	28.5%	-2.0%	29.5%	26.5%	3.1%	31.2%	-1.7%
EPS (NT$)	7.50	4.58	63.6%	7.15	4.8%	6.45	5.56	15.9%	7.26	-11.2%	8.20	6.62	23.9%	8.82	-7.1%
YoY	*124.3%*	37.2%		*112.3%*		*-14.0%*	21.4%		*1.5%*		*27.2%*	19.0%		*21.5%*	

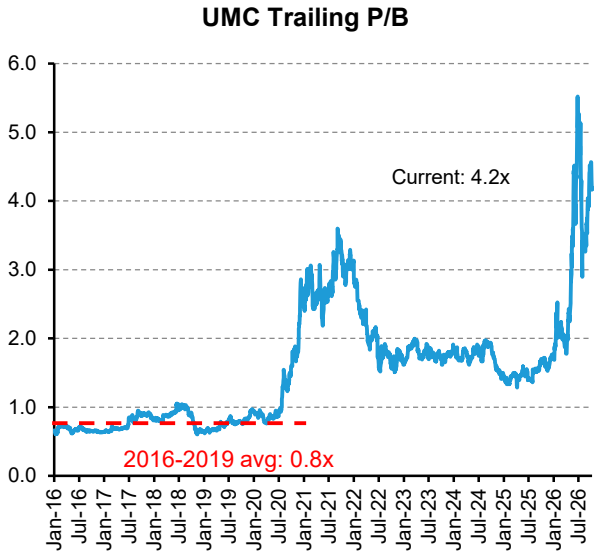
Source: Company reports, Bloomberg, Bernstein estimates and analysis

EXHIBIT 14: **After adjusting for large investment income/ (loss), we model adjusted ROE to improve to c. 14% in 2026, 15% in 2027 & 18% in 2028.**



Source: Company report, Bernstein estimates and analysis

EXHIBIT 15: **UMC is now trading at 4.2x forward P/B.**



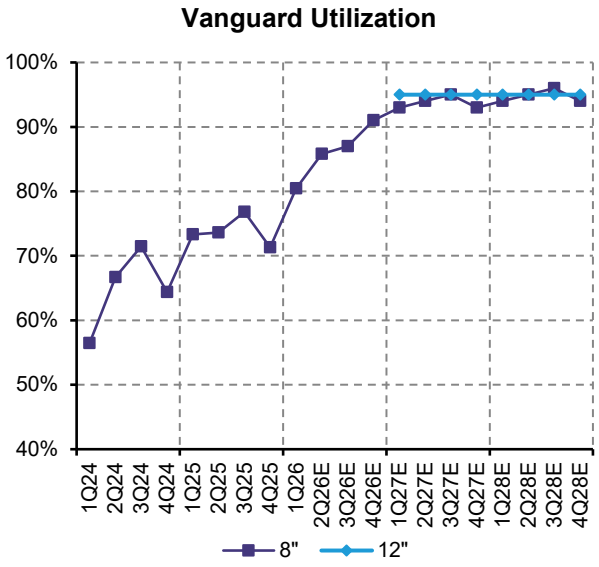
Source: Bloomberg and Bernstein analysis

We're positive on Vanguard but maintain Market-Perform on full valuation.

- The ramp of 12" new fab (VSMC) is better than our expectations. With the interposer orders from TSMC and strong demand for power management products, we now expect close to full utilization for VSMC at the beginning of ramp-up stage, and limited cannibalization of the legacy 8" business. We hence raised our ASP forecasts for 2027 & 28 (Exhibit 16 - Exhibit 17). On the cost side, depreciation from VSMC is expected to kick in from 4Q26 and drag GM by L-MSD (Low-to-Mid Single Digit) % in 2027 and HSD (High-Single-Digit) % in 2028. With better product mix (i.e. more contribution from 12" wafers), we believe the additional depreciation burden can be largely offset and now forecast GM to remain flattish across 2026-2028 (Exhibit 18 - Exhibit 19).

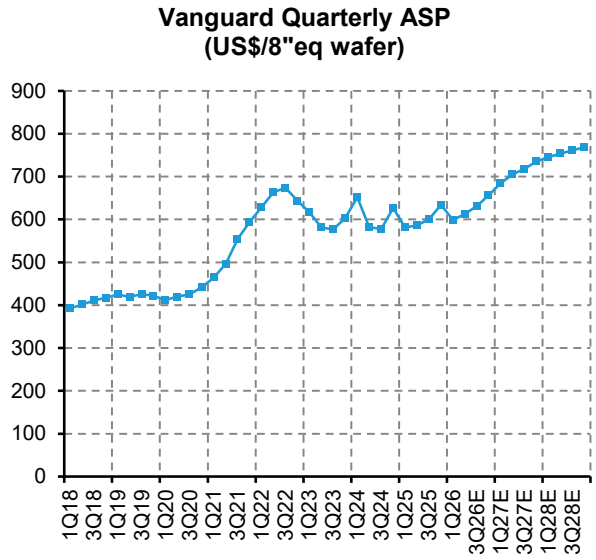
- We raise our target P/B to 5x from 4x on better visibility of VSMC and positive pricing outlook of 8" power management products. We see the possibility of Vanguard adding a phase 2 in Singapore, but choose not to model it until we have better visibility on the project. With higher Q5 BVPS, we increase our price target to NT$194.00/share, implying 2% upside as of October 8 market close. Dividend yield will add c. 2.5% return. Maintain Market-Perform on full valuation (Exhibit 21 - Exhibit 22).

EXHIBIT 16: **We expect VSMC to see close to full loading from the beginning of ramping up, and minimal cannibalization of the legacy 8" business.**



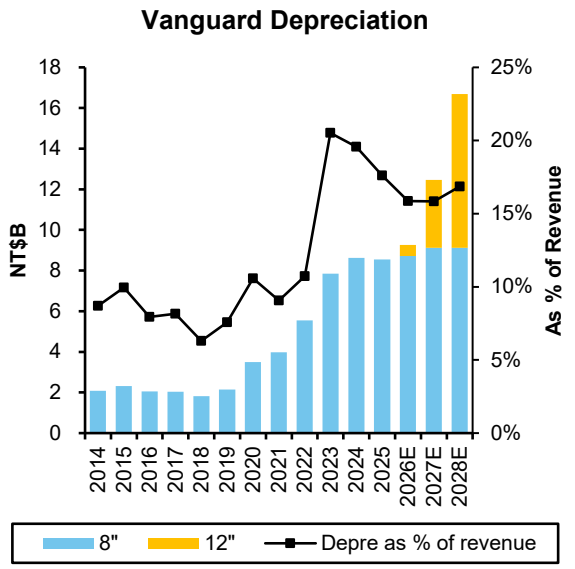
Source: Company reports, Bernstein estimates and analysis

EXHIBIT 17: **We forecast only a moderate price hike within the year but more substantial pricing adjustments later.**



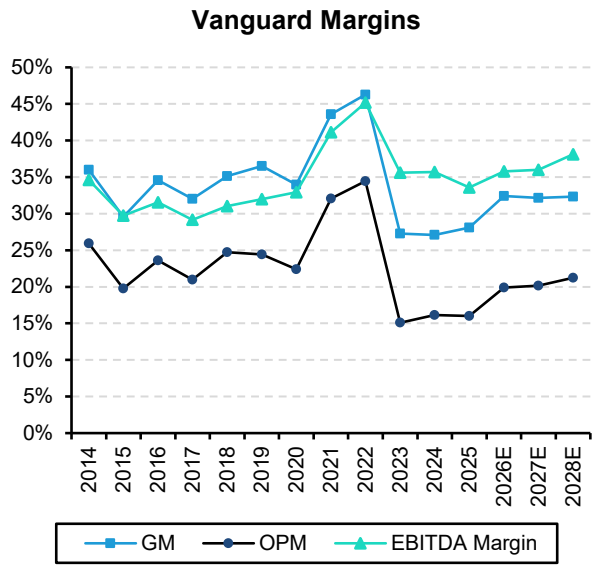
Source: Company reports, Bernstein estimates and analysis

EXHIBIT 18: **Depreciation from VSMC is expected to kick in from 4Q26...**



Source: Company reports, Bernstein estimates and analysis

EXHIBIT 19: **... and drag GM by L-MSD% in 2027 and HSD% in 2028.**



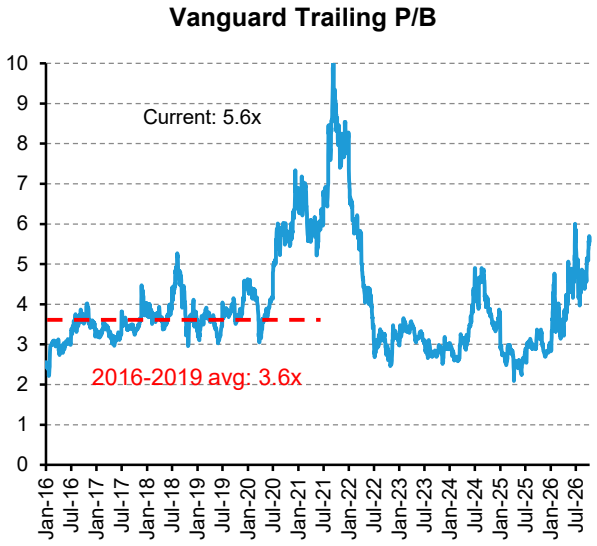
Source: Company reports, Bernstein estimates and analysis

EXHIBIT 20: **We're largely in line with consensus on both topline and bottom line across 2026-28.**

	2026E					2027E					2028E				
	Bernstein					Bernstein					Bernstein				
Vanguard	New	Old	% Chg	Consensus	% Diff	New	Old	% Chg	Consensus	% Diff	New	Old	% Chg	Consensus	% Diff
Revenue (NT$ M)	58,344	59,701	-2.3%	58,364	0.0%	78,642	70,805	11.1%	78,306	0.4%	98,989	88,847	11.4%	99,127	-0.1%
YoY	*20.1%*	*22.9%*		*20.6%*		*34.8%*	*18.6%*		*34.2%*		*25.9%*	*25.5%*		*26.6%*	
Gross Margin	32.4%	32.1%	0.3%	32.3%	0.1%	32.1%	32.0%	0.2%	31.6%	0.5%	32.3%	32.3%	0.0%	31.5%	0.8%
Operating Margin	19.9%	19.4%	0.5%	20.4%	-0.4%	20.2%	18.4%	1.8%	20.5%	-0.3%	21.2%	20.1%	1.1%	22.6%	-1.3%
EPS (NT$)	5.93	5.70	4.1%	6.02	-1.5%	7.76	6.59	17.8%	7.88	-1.5%	10.07	8.95	12.5%	9.66	4.3%
YoY	*37.9%*	*32.5%*		*39.8%*		*31.0%*	*15.7%*		*31.0%*		*29.8%*	*35.9%*		*22.5%*	

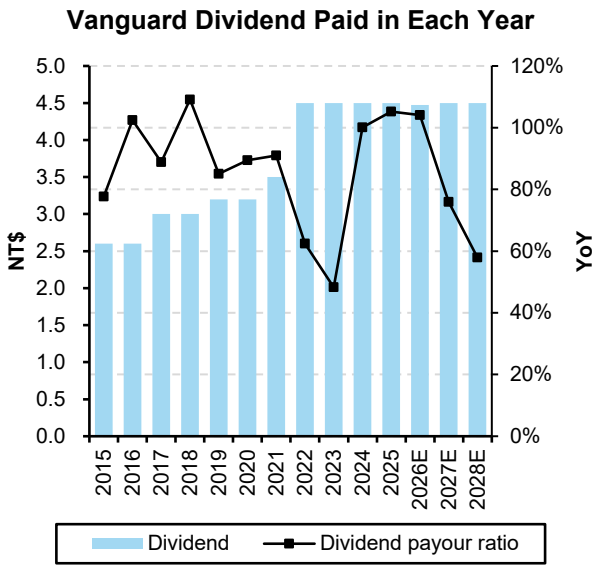
Source: Company reports, Bloomberg, Bernstein estimates and analysis

EXHIBIT 21: **Vanguard is now trading at 5.6x forward P/B, implying 2% potential upside.**



Source: Company reports, Bernstein estimates and analysis

EXHIBIT 22: **We expect Vanguard to keep paying annual dividend of NT$4.5/share before 2029 as guided.**



Source: Company reports, Bernstein estimates and analysis

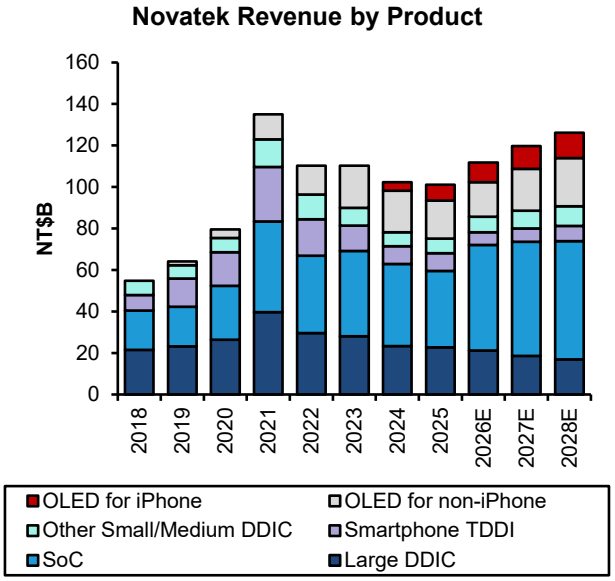
Novatek benefits from Edge AI, but the margin benefit from low-cost memory inventory is transitory. We maintain Market-Perform.

- Novatek gave a solid 3Q26 revenue guidance earlier, driven primarily by strong inventory build-up from smartphone OEMs and continued momentum in smart edge vision-related products. We hence revised up 3Q26 revenue to reflect more resilient-than-expected smartphone demand. Meanwhile, gross margin guidance was also solid, likely supported by better product mix (i.e. higher contribution from SoC products), successful cost pass-through, and low-cost memory inventory. We hence slightly revised up our 3Q26 gross margin.

- On the other hand, we maintained lower-than-consensus gross margin estimates for 2027 & 2028 as we believe the tailwind from cheaper memory inventory will eventually taper off. We also noted that recent strength is partially helped by customer restocking, which, once reaches its target levels, may trigger order cuts.

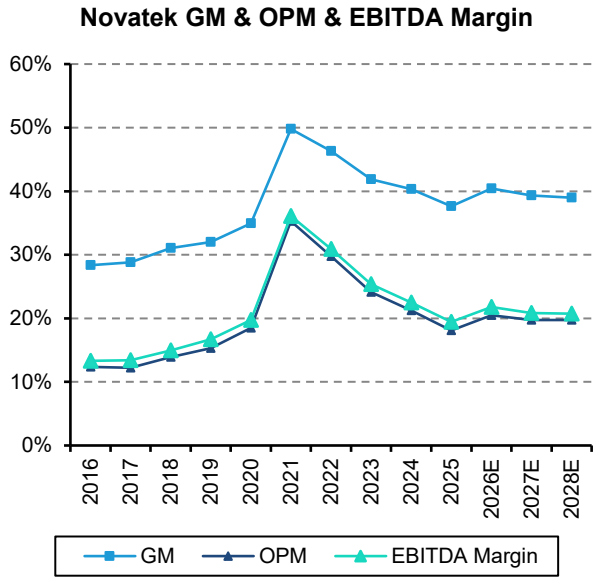
- The stock is now trading at 15.1x forward P/E. We revised up our target price to NT$490.00 from NT$480.00 based on slightly higher Q5-Q8 earnings assumption and the same 14x target P/E multiple. It implies c. 7% downside in stock price. Dividend is likely to provide 4% yield. Maintain Market-Perform (Exhibit 25 - Exhibit 26).

EXHIBIT 23: **We expect smart edge vision to become a key strategic growth area for Novatek.**



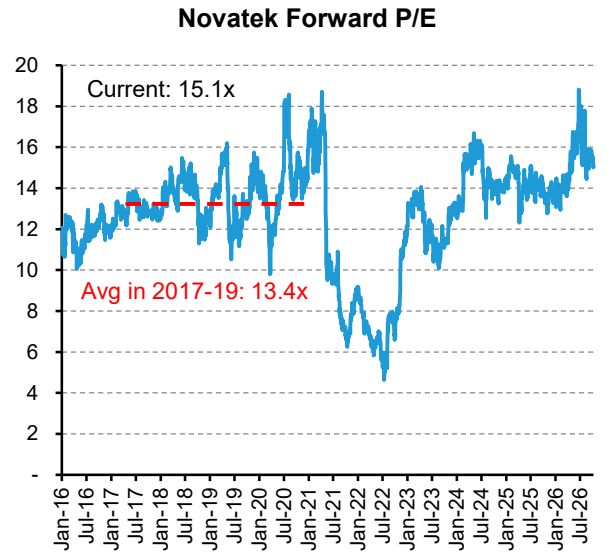
Source: Company reports, Bernstein estimates and analysis

EXHIBIT 24: **GM in 2026 got a temporary boost from low-cost memory inventory. We assume this benefit will taper off in 2027.**



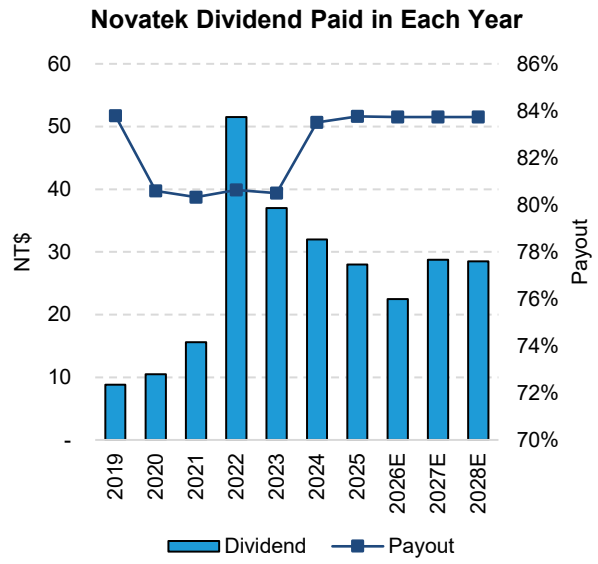
Source: Company reports, Bernstein estimates and analysis

EXHIBIT 25: **The stock is now trading at 15.1x forward P/E, implying 7% potential downside.**



Source: Company reports, Bernstein estimates and analysis

EXHIBIT 26: **A 4% dividend yield should limit the downside.**



Source: Company reports, Bernstein estimates and analysis

EXHIBIT 27: **We're slightly below consensus on 2027 & 28 gross margin as we believe the tailwind from low-cost memory inventory will gradually taper off.**

	2026E					2027E					2028E				
	Bernstein					Bernstein					Bernstein				
Novatek	New	Old	% Chg	Consensus	% Diff	New	Old	% Chg	Consensus	% Diff	New	Old	% Chg	Consensus	% Diff
Revenue (NT$ M)	111,640	110,418	1.1%	106,588	4.7%	119,538	116,348	2.7%	115,219	3.7%	126,051	122,389	3.0%	122,389	3.0%
YoY	*10.9%*	*9.7%*		*6.0%*		*7.1%*	*5.4%*		*8.1%*		*5.4%*	*5.2%*		*6.2%*	
Gross Margin	40.4%	39.0%	1.4%	39.1%	1.3%	39.3%	39.0%	0.4%	38.9%	0.5%	39.0%	39.1%	-0.1%	39.1%	-0.3%
Operating Margin	20.5%	19.0%	1.5%	18.6%	1.9%	19.8%	19.3%	0.4%	19.3%	0.4%	19.8%	19.7%	0.0%	19.7%	0.2%
EPS (NT$)	34.32	30.43	12.8%	28.57	20.1%	34.03	32.54	4.6%	31.54	7.9%	35.87	34.89	2.8%	34.89	2.8%
YoY	*27.7%*	*13.3%*		*9.2%*		*-0.8%*	*6.9%*		*10.4%*		*5.4%*	*7.2%*		*10.6%*	

Source: Company reports, Bloomberg, Bernstein estimates and analysis

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APPENDIX - FINANCIAL FORECASTS

EXHIBIT 28: **UMC Financial Forecasts**

	2026				2027								
NT$ in Millions	Q1	Q2	Q3E	Q4E	Q1E	Q2E	Q3E	Q4E	2024	2025	2026E	2027E	2028E
Income Statement													
Net Sales	61,038	68,733	75,811	78,845	78,488	81,642	85,985	87,644	232,303	237,553	284,427	333,759	382,920
Gross Profit	17,818	22,323	27,006	28,642	27,641	29,146	31,908	32,984	75,654	68,906	95,790	121,680	147,637
SG&A Expense	2,523	3,158	3,327	3,489	3,515	3,620	3,706	3,718	9,749	9,223	12,498	14,558	15,240
R&D Expense	4,575	4,747	5,040	5,288	5,055	5,337	5,504	5,612	15,616	17,725	19,650	21,507	22,484
EBIT	11,276	14,950	19,208	20,494	19,717	20,863	23,421	24,397	51,613	43,949	65,929	88,397	113,040
EBITDA	27,263	31,188	36,094	38,120	37,703	39,377	42,671	43,972	99,780	103,208	132,666	163,723	195,628
Non-operating Income	5,367	30,236	1,365	1,234	1,866	1,948	2,118	1,998	4,607	5,699	38,202	7,930	9,494
Pretax Income	16,644	45,186	20,573	21,728	21,582	22,811	25,539	26,395	56,220	49,648	104,131	96,328	122,534
Tax	527	2,962	3,497	3,694	3,561	3,764	4,214	4,355	9,113	8,113	10,680	15,894	20,218
Net Income	16,117	42,224	17,076	18,034	18,021	19,048	21,325	22,040	47,106	41,535	93,451	80,434	102,316
Earnings Per Share (NT$)	1.29	3.39	1.37	1.45	1.44	1.53	1.71	1.77	3.80	3.34	7.50	6.45	8.20
Margin Analysis													
Gross Profit	29.2%	32.5%	35.6%	36.3%	35.2%	35.7%	37.1%	37.6%	32.6%	29.0%	33.7%	36.5%	38.6%
EBIT	18.5%	21.8%	25.3%	26.0%	25.1%	25.6%	27.2%	27.8%	22.2%	18.5%	23.2%	26.5%	29.5%
EBITDA	44.7%	45.4%	47.6%	48.3%	48.0%	48.2%	49.6%	50.2%	43.0%	43.4%	46.6%	49.1%	51.1%
Pretax Income	27.3%	65.7%	27.1%	27.6%	27.5%	27.9%	29.7%	30.1%	24.2%	20.9%	36.6%	28.9%	32.0%
Net Income	26.4%	61.4%	22.5%	22.9%	23.0%	23.3%	24.8%	25.1%	20.3%	17.5%	32.9%	24.1%	26.7%
Sequential Growth													
Net Sales	-1.2%	12.6%	10.3%	4.0%	-0.5%	4.0%	5.3%	1.9%	4.4%	2.3%	19.7%	17.3%	14.7%
Gross Profit	-6.0%	25.3%	21.0%	6.1%	-3.5%	5.4%	9.5%	3.4%	-2.7%	-8.9%	39.0%	27.0%	21.3%
EBIT	-7.8%	32.6%	28.5%	6.7%	-3.8%	5.8%	12.3%	4.2%	-10.8%	-14.8%	50.0%	34.1%	27.9%
EBITDA	-2.1%	14.4%	15.7%	5.6%	-1.1%	4.4%	8.4%	3.0%	1.4%	3.4%	28.5%	23.4%	19.5%
Pretax	7.4%	171.5%	-54.5%	6%	-1%	6%	12%	3%	-20.7%	-11.7%	109.7%	-7.5%	27.2%
Net Income	61.2%	162.0%	-59.6%	6%	0%	6%	12%	3%	-23.3%	-11.8%	125.0%	-13.9%	27.2%
Earnings Per Share	59.3%	162.8%	-59.6%	5.6%	-0.1%	5.7%	12.0%	3.3%	-23.0%	-12.1%	124.3%	-14.0%	27.2%
Balance Sheet													
Cash and Cash Equivalents									105,000	110,660	204,785	212,420	303,254
Accounts Receivable									33,343	31,274	41,958	46,460	52,282
Inventory									35,782	37,228	44,867	47,619	52,621
Net Fixed Assets									279,059	271,395	267,101	284,575	240,388
Total Assets									570,201	578,996	762,385	797,116	861,393
Accounts Payable									7,633	9,170	10,009	10,840	12,221
Total Debt									81,462	78,970	138,819	139,483	138,369
Total Equity									378,185	379,855	479,034	509,568	561,984
Cash Flow													
Cash Flow From Operations									93,872	99,864	140,029	149,671	180,247
Cash Flow From Investments									(85,941)	(53,154)	(72,627)	(92,800)	(38,400)
Cash Flow From Financing									(39,200)	(39,204)	25,142	(49,237)	(51,014)
Depreciation & Amortization									48,168	59,259	66,737	75,325	82,587
Capex									(88,544)	(47,745)	(63,749)	(92,800)	(38,400)
Key Financial Metrics													
Return on Common Equity									12.8%	11.0%	21.8%	16.3%	19.1%
Return on Assets									8.4%	7.3%	13.9%	10.3%	12.3%
Capex Intensity									38%	20%	22%	28%	10%
Asset Turnover									0.4x	0.4x	0.4x	0.4x	0.4x
Debt / Equity									22%	21%	29%	27%	25%
Net Debt / Equity									-6%	-8%	-14%	-14%	-29%
Debt / Capital									18%	17%	22%	21%	20%
Dividend Per Share (NT$)									3.0	2.9	2.5	4.0	4.0

Source: Company reports, Bernstein analysis and estimates

EXHIBIT 29: **Vanguard Financial Forecasts**

	2026				2027								
NT$ in Millions	**Q1**	**Q2**	**Q3E**	**Q4E**	**Q1E**	**Q2E**	**Q3E**	**Q4E**	**2024**	**2025**	**2026E**	**2027E**	**2028E**
Income Statement													
Net Sales	12,532	14,245	15,092	16,476	17,362	18,902	20,474	21,904	44,055	48,591	58,344	78,642	98,989
Gross Profit	3,671	4,604	5,009	5,616	5,499	6,082	6,598	7,091	11,933	13,654	18,901	25,271	31,981
SG&A Expense	974	1,023	1,117	1,203	1,181	1,248	1,290	1,336	2,740	3,691	4,317	5,054	5,685
R&D Expense	610	680	785	890	990	1,077	1,106	1,183	2,081	2,190	2,965	4,355	5,269
EBIT	2,087	2,901	3,108	3,524	3,329	3,757	4,203	4,572	7,111	7,773	11,619	15,861	21,027
EBITDA	4,278	5,128	5,448	6,022	6,381	6,895	7,319	7,721	15,733	16,326	20,877	28,316	37,719
Non-operating Income	350	412	175	389	373	389	316	370	1,641	1,477	1,327	1,448	1,514
Pretax Income	2,437	3,313	3,283	3,913	3,702	4,146	4,519	4,942	8,752	9,251	12,945	17,310	22,541
Tax	273	420	657	822	629	705	768	840	1,705	1,480	2,171	2,943	3,832
Net Income	2,164	2,894	2,626	3,091	3,073	3,441	3,751	4,102	7,047	7,770	10,775	14,367	18,709
Earnings Per Share (NT$)	1.22	1.62	1.42	1.67	1.66	1.86	2.03	2.22	4.28	4.30	5.93	7.76	10.07
Margin Analysis													
Gross Profit	29.3%	32.3%	33.2%	34.1%	31.7%	32.2%	32.2%	32.4%	27.1%	28.1%	32.4%	32.1%	32.3%
EBIT	16.7%	20.4%	20.6%	21.4%	19.2%	19.9%	20.5%	20.9%	16.1%	16.0%	19.9%	20.2%	21.2%
EBITDA	34.1%	36.0%	36.1%	36.6%	36.8%	36.5%	35.7%	35.2%	35.7%	33.6%	35.8%	36.0%	38.1%
Pretax Income	19.4%	23.3%	21.8%	23.7%	21.3%	21.9%	22.1%	22.6%	19.9%	19.0%	22.2%	22.0%	22.8%
Net Income	17.3%	20.3%	17.4%	18.8%	17.7%	18.2%	18.3%	18.7%	16.0%	16.0%	18.5%	18.3%	18.9%
Sequential Growth													
Net Sales	-0.5%	13.7%	5.9%	9.2%	5.4%	8.9%	8.3%	7.0%	15.1%	10.3%	20.1%	34.8%	25.9%
Gross Profit	6.0%	25.4%	8.8%	12.1%	-2.1%	10.6%	8.5%	7.5%	14.2%	14.4%	38.4%	33.7%	26.6%
EBIT	15.5%	39.0%	7.1%	13.4%	-5.5%	12.8%	11.9%	8.8%	23.2%	9.3%	49.5%	36.5%	32.6%
EBITDA	6.4%	19.9%	6.2%	10.5%	6.0%	8.1%	6.1%	5.5%	15.5%	3.8%	27.9%	35.6%	33.2%
Pretax	16.3%	36.0%	-0.9%	19.2%	-5.4%	12.0%	9.0%	9.4%	-0.9%	5.7%	39.9%	33.7%	30.2%
Net Income	29.8%	33.7%	-9.2%	17.7%	-0.6%	12.0%	9.0%	9.4%	-4.4%	10.3%	38.7%	33.3%	30.2%
Earnings Per Share	28.5%	32.4%	-12.3%	17.7%	-0.4%	12.0%	9.0%	9.3%	-4.9%	0.6%	37.9%	31.0%	29.8%
Balance Sheet													
Cash and Cash Equivalents									60,695	39,635	81,548	95,217	122,315
Accounts Receivable									6,235	6,748	9,578	11,316	14,435
Inventory									6,221	6,937	8,735	11,271	13,342
Net Fixed Assets									44,141	113,869	94,938	81,831	53,138
Total Assets									148,706	200,469	226,100	234,759	241,432
Accounts Payable									1,416	1,724	2,244	3,220	3,812
Total Debt									24,656	31,939	37,913	30,665	27,144
Total Equity									68,667	83,590	108,309	133,426	149,890
Cash Flow													
Cash Flow From Operations									24,731	25,439	23,069	13,391	24,533
Cash Flow From Investments									(11,699)	(59,711)	(3,561)	775	12,120
Cash Flow From Financing									17,530	13,460	22,357	(497)	(9,555)
Depreciation & Amortization									8,622	8,552	9,258	12,455	16,693
Capex									(15,909)	(63,265)	(63,700)	(29,348)	(8,000)
Key Financial Metrics													
Return on Common Equity									12.8%	12.4%	16.6%	20.3%	23.6%
Return on Assets									5.6%	4.5%	5.2%	6.3%	7.9%
Capex Intensity									36.1%	130.2%	109.2%	37.3%	8.1%
Asset Turnover									0.3x	0.2x	0.3x	0.3x	0.4x
Debt / Equity									35.9%	38.2%	35.0%	23.0%	18.1%
Net Debt / Equity									-52%	-9%	-40%	-48%	-63%
Debt / Capital									26.4%	27.6%	25.9%	18.7%	15.3%
Dividend Per Share (NT$)									4.5	4.5	4.5	4.5	4.5

Source: Company reports, Bernstein analysis and estimates

EXHIBIT 30: **Novatek Financial Forecasts**

	2026				2027								
NT$ in Millions	**Q1**	**Q2**	**Q3E**	**Q4E**	**Q1E**	**Q2E**	**Q3E**	**Q4E**	**2024**	**2025**	**2026E**	**2027E**	**2028E**
Income Statement													
Net Sales	23,145	28,660	30,776	29,059	26,631	29,332	31,070	32,505	102,788	100,663	111,640	119,538	126,051
Gross Profit	9,040	12,157	12,485	11,430	10,525	11,509	12,181	12,796	41,472	37,908	45,112	47,011	49,123
SG&A Expense	736	841	862	828	879	880	870	845	2,859	2,946	3,267	3,474	3,583
R&D Expense	4,276	4,850	5,046	4,779	4,731	5,041	5,174	4,979	16,807	16,725	18,950	19,926	20,642
EBIT	4,028	6,466	6,577	5,823	4,915	5,588	6,136	6,972	21,806	18,237	22,894	23,611	24,898
EBITDA	4,389	6,827	6,940	6,177	5,264	5,924	6,455	7,293	23,120	19,573	24,333	24,935	26,149
Non-operating Income	362	1,316	266	253	288	311	336	256	2,320	1,296	2,196	1,192	1,239
Pretax Income	4,389	7,782	6,843	6,076	5,203	5,899	6,472	7,228	24,126	19,534	25,090	24,803	26,137
Tax	622	1,454	1,129	1,003	859	973	1,068	1,193	3,784	3,186	4,208	4,092	4,313
Net Income	3,767	6,328	5,714	5,073	4,345	4,926	5,405	6,035	20,342	16,348	20,882	20,710	21,824
Earnings Per Share (NT$)	6.19	10.40	9.39	8.34	7.14	8.10	8.88	9.92	33.43	26.87	34.32	34.03	35.87
Margin Analysis													
Gross Profit	39.1%	42.4%	40.6%	39.3%	39.5%	39.2%	39.2%	39.4%	40.3%	37.7%	40.4%	39.3%	39.0%
EBIT	17.4%	22.6%	21.4%	20.0%	18.5%	19.1%	19.8%	21.4%	21.2%	18.1%	20.5%	19.8%	19.8%
EBITDA	19.0%	23.8%	22.6%	21.3%	19.8%	20.2%	20.8%	22.4%	22.5%	19.4%	21.8%	20.9%	20.7%
Pretax Income	19.0%	27.2%	22.2%	20.9%	19.5%	20.1%	20.8%	22.2%	23.5%	19.4%	22.5%	20.7%	20.7%
Net Income	16.3%	22.1%	18.6%	17.5%	16.3%	16.8%	17.4%	18.6%	19.8%	16.2%	18.7%	17.3%	17.3%
SG&A as % of Sales	3.2%	2.9%	2.8%	2.9%	3.3%	3.0%	2.8%	2.6%	2.8%	2.9%	2.9%	2.9%	2.8%
R&D as % of Sales	18.5%	16.9%	16.4%	16.4%	17.8%	17.2%	16.7%	15.3%	16.4%	16.6%	17.0%	16.7%	16.4%
Sequential Growth													
Net Sales	1.4%	23.8%	7.4%	-5.6%	-8.4%	10.1%	5.9%	4.6%	-6.9%	-2.1%	10.9%	7.1%	5.4%
Gross Profit	3.8%	34.5%	2.7%	-8.5%	-7.9%	9.3%	5.8%	5.1%	-10.3%	-8.6%	19.0%	4.2%	4.5%
EBIT	4.3%	60.5%	1.7%	-11.5%	-15.6%	13.7%	9.8%	13.6%	-18.2%	-16.4%	25.5%	3.1%	5.5%
EBITDA	4.0%	55.5%	1.7%	-11.0%	-14.8%	12.5%	9.0%	13.0%	-17.4%	-15.3%	24.3%	2.5%	4.9%
Pretax	-0.6%	77.3%	-12.1%	-11.2%	-14.4%	13.4%	9.7%	11.7%	-14.5%	-19.0%	28.4%	-1.1%	5.4%
Net Income	2.1%	68.0%	-9.7%	-11.2%	-14.4%	13.4%	9.7%	11.7%	-12.8%	-19.6%	27.7%	-0.8%	5.4%
Earnings Per Share	2.1%	68.0%	-9.7%	-11.2%	-14.4%	13.4%	9.7%	11.7%	-12.8%	-19.6%	27.7%	-0.8%	5.4%
Balance Sheet													
Cash and Cash Equivalents									48,771	42,419	56,859	57,394	60,282
Accounts Receivable									20,274	21,099	22,771	27,205	29,864
Inventory									8,701	8,641	10,108	11,532	12,667
Net Fixed Assets									7,023	7,210	2,902	2,415	2,047
Total Assets									102,401	97,366	116,648	122,710	129,240
Accounts Payable									10,635	9,352	12,638	13,637	14,819
Total Debt									455	552	570	622	652
Total Equity									67,923	67,739	79,748	82,972	87,453
Cash Flow													
Cash Flow From Operations									16,652	13,629	27,061	18,807	21,083
Cash Flow From Investments									(849)	(1,232)	3,377	(837)	(882)
Cash Flow From Financing									(20,956)	(18,741)	(16,023)	(17,435)	(17,313)
Depreciation & Amortization									1,314	1,336	1,439	1,324	1,251
Capex									(1,276)	(1,211)	3,014	(837)	(882)
Key Financial Metrics													
Return on Common Equity									30.1%	24.1%	28.3%	25.5%	25.6%
Return on Assets									19.7%	16.4%	19.5%	17.3%	17.3%
Debt / Equity									0.7%	0.8%	0.7%	0.7%	0.7%
Net Debt / Equity									-71%	-62%	-71%	-68%	-68%
Debt / Capital									0.7%	0.8%	0.7%	0.7%	0.7%
Dividend Per Share (NT$)									32.0	28.0	22.5	28.7	28.5

Source: Company reports, Bernstein analysis and estimates

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